

Paper III v1.3 Internal Audit Final Report

Internal author-side audit — not independent

22 August 2026 — Overall verdict: PASS

Protocol: INTERNAL_AUDIT_STANDARD_v1.3.

Release position: candidate for Paper III’s first formal public release.

Lean execution: recorded consolidated build reviewed; full build not rerun during the audit.

1 Frozen target

Formal archive SHA-256:

2eb0ff20a9dae6610a46026355374570d5afdf6ea89837ea7f9dd29da10b9d300.

Toolchain: Lean 4.28.0; Mathlib v4.28.0 at commit 8f9d9cff6bd728b17a24e163c9402775d9e6a365.

Recorded build: exit 0; 8,719 successful jobs. Eight axiom-query runs record exactly `[propext, Classical.choice, Quot.sound]` on every queried canonical surface. The archive and both manifests verify.

2 Gate results

Gate	Verdict	Principal evidence
G0	PASS	Six artifacts, LF sidecar, manifests, archive hash and CRC.
G1	PASS	Main bound, sharpness, three regimes and exact EN/ES scope.
G2	PASS	12 identities; 351 LP/exact cases; 78,384 margins; 372 ILPs; 360 packing/corridor cases.
G3	PASS	Canonical model bridges, AX1/AX2, induction and final theorem.
G4	PASS build	recorded 8,719-job build and eight axiom-query runs reviewed.
G5	PASS	31 consistency checks; 144/144 headings; notation and citations aligned.
G6	PASS	EN 45 / ES 46 pages; embedded fonts; all 91 pages inspected.
G7	PASS internal	Bounded prior-art conclusion; independent novelty gate open.

G8

PASS

Clean public freeze and delivery tree; no prior-version dependency.

The executable suite reports **133/133 checks passed** and every gate G0–G8 as PASS.

3 External-finding regression

The v1.3 target closes the previous wording, notation, citation, stale-label and formal-interface findings internally. It adds two-sided canonical bridges between the paper and Yuster-style fractional-packing models and explicitly queries the AX1/AX2 and K-EPS, K-CORRIDOR, K-SPARSE, K-COVER and K-GLOBAL surfaces. Two disclosed legacy comparison-route axioms are unimported and absent from all canonical footprints.

This is internal formal-coverage closure, not independent mathematical rederivation. External review must independently check bridge semantics, quantitative tolerances, the hypergraph-to-graph passage, the Sections 4–9 analytic core, dependency closure, citations and novelty.

4 Overall verdict

PASS. No unresolved internal blocker or major finding remains. The target is ready for a new external adversarial audit; all external review and release gates remain open.